

The Resilient, but Limited Effect of Process on Fairness and Legitimacy: Evidence from Wind Energy Siting

Michael Hankinson*

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Abstract

Do people care how a decision was made or only what was decided? Most research uses either retrospective evaluations or experiments that announce the decision before asking about the process. Both cannot say whether procedural values hold independent of outcomes. In a preregistered survey experiment on wind energy siting among rural Americans, I independently randomize who decides, how the public participates, whether the project is approved or denied, and whether respondents learn that outcome before evaluating the process. Before the outcome is known, both centralized decision-making and limited public input decrease perceived fairness and legitimacy. These effects persist once the decision is announced. Still, receiving one's preferred outcome has a much larger effect on ex post evaluations compared to process alone. Methodologically, these findings underscore the risk of retrospective evaluations common in fieldwork of siting conflicts. Regarding governance, they offer paths for maintaining legitimacy while reforming energy siting.

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*Assistant Professor, Department of Political Science, George Washington University.

In 2009, Norway replaced local control over wind energy permitting with a centralized, national framework. Although turbines were built more quickly, subsequent local protests pressured the government to repeal the reform in 2023 (Vasstrøm et al. 2024). Today, a similar story is unfolding across North America and Europe. States and provinces are preempting local control over siting decisions to expedite energy construction (Chaudhuri and Susskind 2024), while residents object not only to the projects but to the fact that the decision is no longer theirs to make (Stokes et al. 2023). But preemption of local control is typically associated with a specific policy outcome—more energy infrastructure. Thus, it is unclear: do people value the process by which a decision gets made (e.g., local control) or do they only value the policy outcome (e.g., blocking proposed infrastructure)?

Political science has good reason to expect both. On one hand, a deep literature argues that people who believe that a decision-making process was fair are more willing to accept decisions with which they disagree (Lind and Tyler 1988; Tyler 1994). Put differently, losers consent because the process was legitimate, not because they got what they wanted (Anderson 2005). On the other hand, these procedural commitments are quickly abandoned when inconvenient (Prothro and Grigg 1960). Experimental research finds that when people evaluate a process after the result is known, they mostly respond to whether they won (Esaiasson et al. 2016).

Adjudicating between these theories is complicated because most evidence on procedural fairness is observational, collected through retrospective evaluations once the structure is built. When a resident near a new energy facility says that the siting process was unfair, we cannot separate a judgment about the process (hearings, public outreach, etc.) from disappointment about the policy outcome (that the facility was built). In contrast, experiments that randomize the process but leave the outcome unstated have the opposite problem (e.g., Liu et al. 2020). A process that sounds more inclusive of residents' opinions also sounds more likely to produce the outcome that respondent prefers. Thus, process and expected outcome move together (Persson, Esaiasson and Gilljam 2013).

To isolate the role of process in siting, I test how energy siting reforms affect perceived fairness and legitimacy via a preregistered survey experiment fielded among residents of rural U.S. ZIP codes, those most likely to encounter a utility-scale wind proposal.¹ Local government is the ideal context for this study because it is where the democratic process is most directly felt by the average voter. Decisions, including those about energy siting, are made by elected officials in open meetings and residents can attend and be heard by the officials casting the votes (Einstein, Palmer and Glick 2019; Sahn 2025). In my experiment, respondents evaluated a hypothetical wind farm under randomly assigned institutional conditions that varied who makes the decision—locally elected officials or a state-appointed board of experts—and how the public participates—in-person meetings or written comment only, both real institutions in government decision-making across North America and Europe (Section A). Crucially, I also randomized whether the project was approved or denied and whether respondents learned that outcome before they evaluated the process.

The outcomes of this study were inspired by *procedural fairness theory* (Tyler 1994). In theory, once a decision is made, an observer evaluates the fairness of the decision-making process. The observer’s perception of procedural fairness is then combined with their predisposition towards the outcome (e.g., the wind farm), leading to either acceptance or rejection of the decision—otherwise known as legitimacy. Thus, prior to the treatments, I asked each respondent’s support for the proposed wind facility. Following the treatments, I asked their perceived fairness of the process and captured their assessment of the outcome’s legitimacy via an original four-item scale.

This study makes three contributions to our understanding of process, legitimacy, and political conflict. First, both centralized decision-making and limited public input reduce perceived fairness and legitimacy, as theorized by the observational literature (Firestone et al. 2020). Second, these effects are resilient to learning the policy outcome. Three of four tests remain significant once the outcome is known and all remain around a percentage point of

¹Anonymized preregistration:
2f990e932e804545bdcf4545780c0a76

https://osf.io/v3y8q/overview?view_only=

their original magnitudes. Third, the effect of process is comparatively limited. Receiving one’s preferred outcome increases perceived fairness and legitimacy by two to six times the magnitude of any process effect.

These results offer insights for researchers and policymakers. Methodologically, evaluations of fairness and legitimacy that occur after the outcome is announced face outcome bias. Thus, researchers should couple fieldwork with experimental, quasi-experimental, and panel designs which can better separate process from outcome. As for governments centralizing siting authority to expedite energy expansion and/or decarbonization, the loss of legitimacy from ending local control can be partly offset by preserving face-to-face public input.²

Design

The experiment was fielded in October and November 2025 using PureSpectrum, allowing for targeted sampling of respondents living in rural areas, who are most likely to encounter large-scale wind energy projects.³ I sampled from rural ZIP codes, those with a RUCA code of 7 or higher (e.g., Nemerever and Rogers 2021). Demographic target quotas for income, race, homeownership, age, gender, and education were based on the 2019–2023 5-year ACS estimates for these rural ZCTAs. The sample contains adults who passed both pretreatment attention checks and answered the fairness outcome, yielding 3,076 respondents. See Section D for descriptive statistics.

Respondents were asked to evaluate a hypothetical wind farm proposed for their community. They were explicitly told that this was not a real proposal, but to thoughtfully consider it as if it were real. The prompt includes both arguments for and against the wind farm, as well as a visualization of the scale of the turbines. Below is the prompt used:

²The author used Claude Code (Anthropic) to write and verify the analysis code that produces the figures and tables reported here and for copyediting. All results were verified by the author, who is responsible for their validity.

³Prior to data collection, I preregistered two experimental designs. The other design was a pilot and bundled the level of government with the level of public input into a single treatment, and thus could not separate the two procedural features. Both experiments were conducted in compliance with relevant laws and were approved by [REDACTED] Institutional Review Boards.

A wind energy developer is proposing to build a wind farm on land in your community. They want to build a farm of 60 turbines, each around 340 feet tall, which would power 85,000 homes per year.⁴ You would be able to see the wind turbines from your neighborhood.

For context, here's a visual of the turbine height (far right) compared to other known structures: [*visual shown to respondents reproduced in Section B*]

Your community is divided. On one hand, some residents are concerned that the facility's size would harm local property values and affect migrating birds. Residents are also concerned about the aesthetics, landscape change, and potential negative impacts on quality of life due to the turbines. Economists have shown that tall turbines like the ones proposed can lower property values by around 5%.

On the other hand, technical reports estimate that the wind farm would bring over \$2 million per year in new tax revenue for local schools and services and reduce greenhouse gas emissions. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers.

Respondents were randomized into one of two branches (Figure 1). In Branch 1, respondents viewed only the process but they were not told the outcome of the siting decision. In Branch 2, respondents viewed both the process and the outcome. I also randomized the level of local control and the level of public input, independently of each other. For level of government, the conditions were:

- “your locally elected legislators”
- “a state-appointed board of energy experts”

For public input, the conditions were:

- “hold a series of local public meetings where residents like you can share your opinions face-to-face with the decision makers”

⁴The energy production estimate was based on reporting of the 60 turbine Honey Creek project. However, the turbine height at Honey Creek is taller than the 340 feet described. In Section C, I discuss the implications of this mismatch for my findings.

- “set up an email account where residents like you can send your thoughts, but there will be no public meetings about the proposal”

Finally, I randomized whether the proposed project would be approved or denied by its respective decision-making body. This outcome was only visible for respondents in Branch 2. The full survey instrument is in Section B.

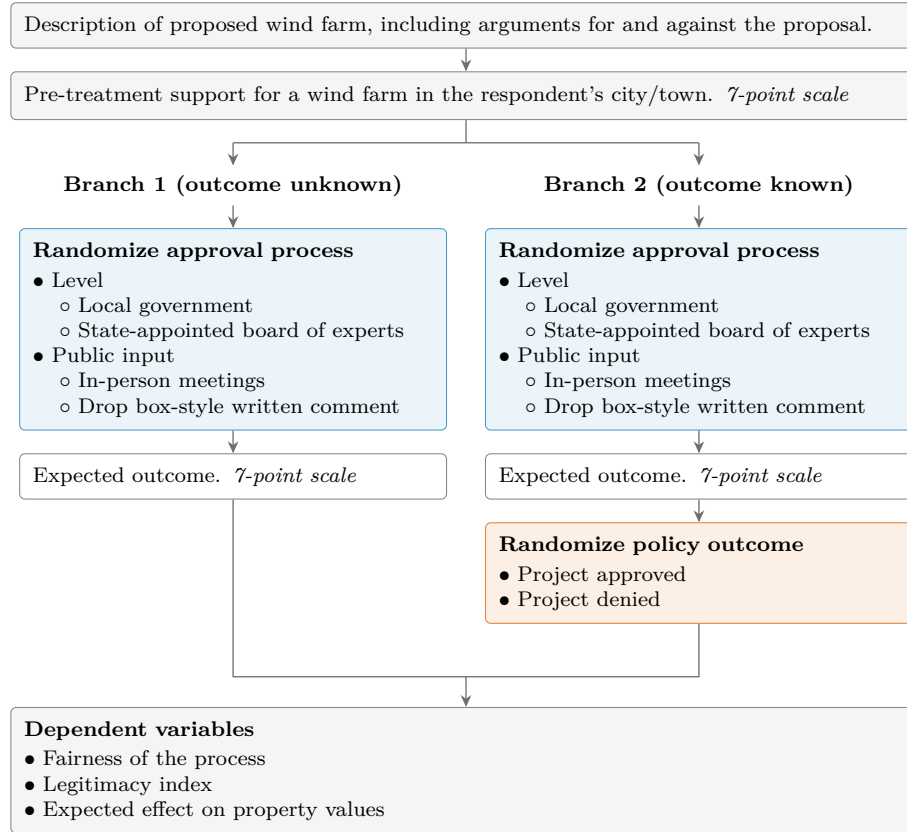


Figure 1: Survey flow.

The primary outcomes of interest are the fairness of the wind turbine siting process and the legitimacy of the final decision. For fairness, the survey asks: “Regardless of what you think about the [eventual] decision, do you believe that a vote by [institution]—after gathering input from residents through [public meetings / email]—is a fair way to decide on the approval of new wind energy farms?” Respondents indicated their perceived fairness on a 7-point scale from “very unfair” to “very fair.” I operationalize this outcome on a continuous scale from 0 to 1 by coding the 7 points from 0 to 6 and dividing by 6.

For legitimacy, I use an original four item scale with the order randomized: “Thinking about the process and [eventual] decision, do you agree or disagree with each of the following statements?”

- “This decision ought to be accepted and considered the final word on the matter.”
- “The body that made this decision had the right to do so.”
- “There ought to be an effort to challenge the decision and get it changed.” (*reversed*)
- “Even if I personally disagree, I respect the outcome of this process.”

Respondents stated their agreement with each statement on a 5-point scale from “strongly agree” to “strongly disagree.” I code the 5 points from 0 to 4 and dividing by 4 for a continuous scale. I then average the four items to create my legitimacy index. The four items are internally consistent with a Cronbach’s alpha of .70. See Section E for validation of the legitimacy scale. All models used OLS regressions with robust standard errors, including covariates for race, gender, education, age, income, homeownership, and partisanship.

Prior to fielding, I preregistered four hypotheses. H1a and H1b posit that in-person meetings and locally elected officials as decision-makers each raise fairness and legitimacy relative to written comment and a state-appointed board of experts. H1c holds that public input is the larger of the two. H2 states that these effects remain significant once the outcome is known. H3 theorizes that the state board penalty is concentrated among respondents who side with their local over their state government. H4 states that the treatment effects of public input and centralization of decision-making are independent of whether the outcome matches the respondent’s preference.

Results

Descriptively, Figure 2 reports the unadjusted means of both outcomes in each experimental condition. There are two takeaways. First, this hypothetical proposal has neutral to positive levels of fairness and legitimacy, averaging .55 for fairness and .51 for legitimacy on a 0-to-1

scale. Thus, the experiment is unlikely to suffer from floor or ceiling effects. Second, we can already see the comparatively limited effect of process next to outcome favorability. While there is variation across the two process dimensions, the highest and lowest levels of our outcomes are found in subgroups based on whether the respondent received their preferred outcome.

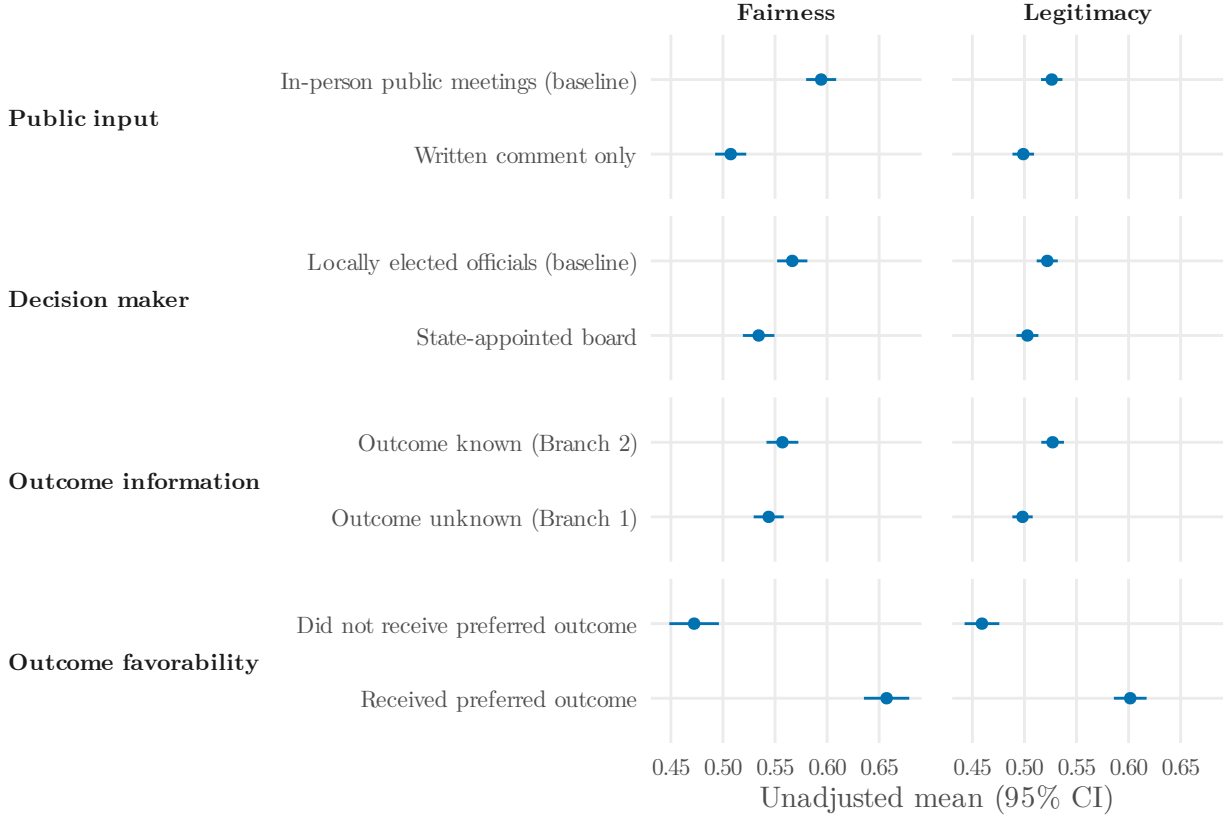


Figure 2: Unadjusted means of fairness and legitimacy by experimental condition on a continuous scale from 0 to 1, shown with 95% confidence intervals.

Effect of Process When Outcome Is Unknown Turning to treatment effects, Figure 3 shows the effect of process on the fairness and legitimacy of the turbine siting among respondents who do not know the outcome (Branch 1 only). In each model, the baseline is the more participatory process: locally elected officials for the level of government and in-person public meetings for the level of input.⁵ Holding no public meetings and only relying on written

⁵See Section H for results in tabular form.

public comment decreases fairness by 9 percentage points ($d = 0.29$), compared to holding in-person meetings (H1a). Having the decision made by a state-appointed board decreases fairness by 3 percentage points ($d = 0.10$), compared to a decision by local elected officials (H1b). As posited in H3, this effect is stronger among those who express more agreement with their local government compared to their state government (Section F).

The effect of process extends to outcome legitimacy. Having the decision made by a state-appointed board of experts decreases legitimacy by 2 percentage points ($d = 0.12$) and holding no public meetings decreases legitimacy by 3 percentage points ($d = 0.14$). Consistent with H1c, the effect of public input is the larger of the two ($p < .01$ by Wald test), however the two effects on legitimacy are indistinguishable. That the effect of process on fairness is greater than on legitimacy is unsurprising, given legitimacy is a combination of one's baseline support for the outcome and evaluation of procedural fairness (Tyler 1994).

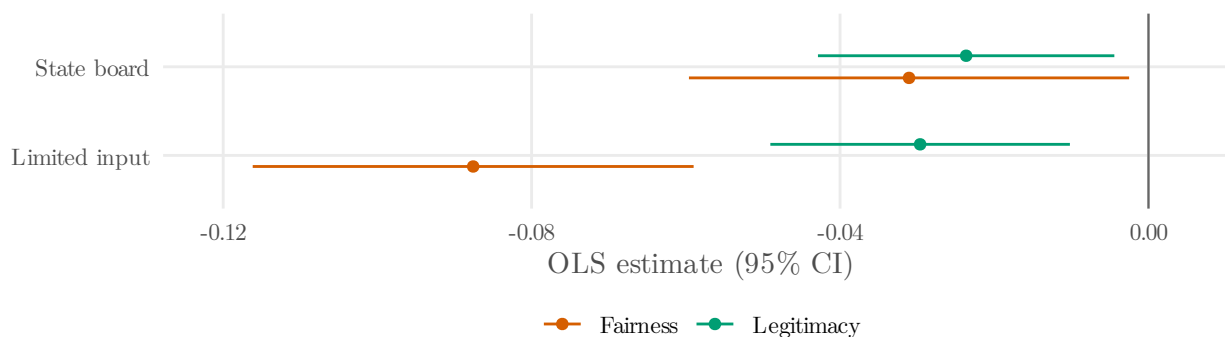


Figure 3: Effect of process when outcome is unknown (Branch 1 only). Baseline is locally elected officials for “State board” and in-person public meetings for “Limited input.” Point estimates shown with 95% confidence intervals.

One possibility is that these effects of process are strategic, with respondents evaluating fairness and legitimacy more favorably for the institutions that they believe will lead to their preferred outcome. Section G shows that these processes do not affect average expectations of project approval, but an exploratory analysis suggests that limited public input decreases expectations of getting what one wants. While this is a potential explanation for Figure 3, the comparison between Branch 1 and 2 below addresses it. In Branch 2, the outcome is revealed, leaving nothing left to expect. If the effect of process does not attenuate, then

these effects cannot be explained by outcome expectations alone.

Resilience of Effects When Outcome Is Known Having established that process affects fairness and legitimacy absent knowledge of the outcome, I test whether people are similarly responsive to process when the outcome is known, reflecting stable average procedural values. Figure 4 shows the independent effects across branches and their difference in the form of an interaction. Across both experiments, knowing the outcome does not alter the effect of process on fairness and legitimacy. Whether or not the outcome is known, respondents still believe that the decision-making process is less fair and less legitimate when the decision is made by the state-appointed board or with limited public input (H2).

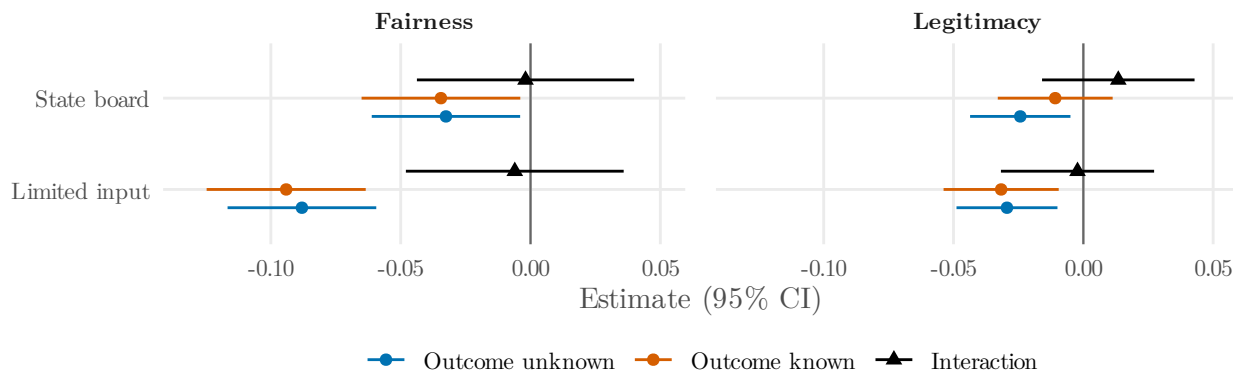


Figure 4: Effect of process by respondent knowledge of the outcome (Branches 1 and 2). Interaction is the difference in effects between the two branches. The baseline is locally elected officials for “State board” and in-person public meetings for “Limited input.” Point estimates shown with 95% confidence intervals.

One potential concern is that the null interactions in Figure 4 reflect insufficient power rather than the resilience of process effects. Section I addresses this concern with an exploratory, one-sided test against the full-strategic prediction: that the interaction equals or exceeds the main process effect in magnitude. Three of four models reject this null at conventional levels. The one exception is the state board effect on legitimacy, the smallest effect in the study. Still, the consistent pattern of near zero interactions across all models is consistent with stable procedural values on average.

Limited Effects Compared to Preferred Outcomes While Figures 4 and 5 show a resilient effect of process on fairness and legitimacy, it is important to contextualize its magnitude. If ex post evaluations of legitimacy are mainly based on whether people received their preferred outcome, then process effects will be difficult to detect through retrospective evaluations.

To contextualize these effects, I regress fairness and legitimacy on an indicator for whether the outcome matches the respondent’s pretreatment preference for the wind farm, dropping those who felt neutral about the proposal. Because this preference is measured before randomization and the outcome is randomized, this match indicator is randomly assigned conditional on preference. Thus, the estimate is a causal effect within preference, pooling the effect of approval among supporters with the effect of denial among opponents. Receiving one’s preferred outcome raises perceived fairness by 18 percentage points and legitimacy by 14 percentage points, multiple times more than any single process effect.

Furthermore, while outcome knowledge in general does not alter the average effect of process (Figure 4), I find evidence that receiving one’s preferred outcome may alter the effect of the decision-making body (H4).⁶ Figure 5 shows that, when it comes to fairness, winners and losers polarize in their responsiveness to who made the decision. Losers care great deal while winners express a null. Notably, such polarization does not occur for the level of public input, where there is a consistent penalty for a lack of face-to-face meetings. Tests of polarization for legitimacy are underpowered.

⁶The pre-analysis plan specified this test within Branch 2 alone, contrasting respondents whose outcome matched their preference against those whose outcome did not. I depart from that specification to retain the outcome-unknown benchmark. Instead, I estimate a single pooled model across both branches in which the process treatments are interacted with a three-level indicator—outcome unknown (Branch 1), outcome known and preferred, and outcome known and not preferred (Branch 2). The preregistered Branch 2 specification yields the same substantive pattern.

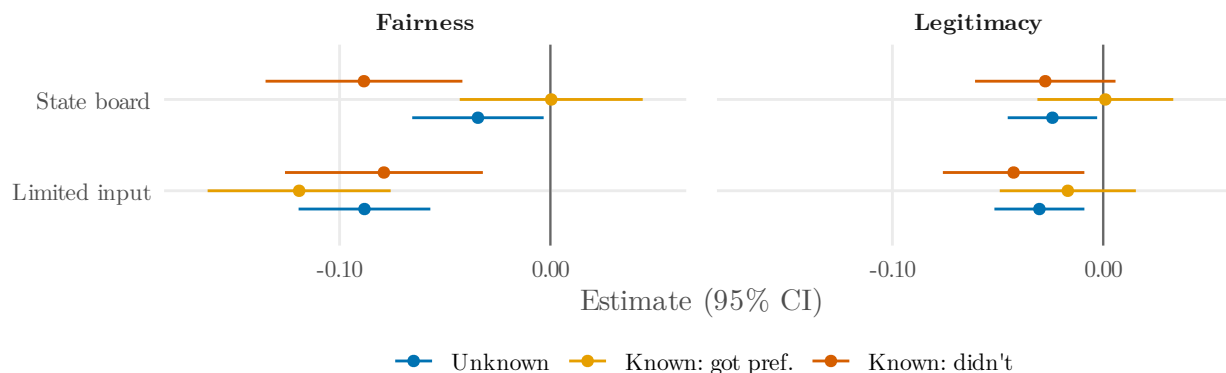


Figure 5: Effect of process within each level of outcome knowledge: outcome unknown (Branch 1), and, among those told the outcome (Branch 2), whether it did or did not match the respondent’s pretreatment preference. The baseline is locally elected officials for “State board” and in-person public meetings for “Limited input.” Point estimates shown with 95% confidence intervals.

Discussion

Face-to-face public input and, to a lesser extent, local control increase perceived fairness and legitimacy. These effects are resilient to knowing whether the institution will approve or deny the proposal. Ex ante procedural evaluations are therefore not merely strategic projections of anticipated outcomes, but reflect stable attitudes toward how decisions ought to be made, at least within the population as a whole. While there is evidence that effects are less stable or perhaps outcome-dependent for the level of decision-making, one interpretation is that winners can discount an ambiguous feature like who decided, whereas public input is solidified as uniformly fair. Still, in retrospective evaluations (Branch 2), process matters much less than getting one’s preferred outcome. This outcome bias should push researchers to move beyond retrospective observational evaluations and pursue more experimental or at least panel designs with repeat sampling throughout the decision-making process (e.g., Firestone et al. 2020).

Another contribution of this study is an experimental framework which isolates whether stable procedural values exist independent of outcome knowledge. To my knowledge, this is the first study to combine the randomization of process, outcomes, and knowledge of outcome

in a manner which captures the full flow of procedural fairness theory. Of course, variation in process can generate a host of downstream mechanisms unaccounted for in this experiment, such as changes in media coverage and the type of public testimony. Likewise, a nearby resident's participation in the process can include hours of meetings with decision-makers, something not easily replicated in an experiment. Consequently, this study's treatment is more similar to how a resident would hear about the permitting process via local media, an experience better reflecting that of the average voter in the community. In short, these findings may either understate the loss of legitimacy if unaccounted for mechanisms amplify procedural concerns or overstate them if the average voter thinks little about how local decisions are made outside of a survey context.

As applied to governance, state and national governments across North America and Europe are increasingly preempting local control to facilitate the supply of new renewable energy. My findings indicate that governments can offset some of the loss in fairness and legitimacy from preemption by maintaining high levels of face-to-face public input. But there remain two limitations. First, just the siting of more energy infrastructure is likely to drive perceptions of fairness to a greater extent than the process alone. Thus, more infrastructure will lead to increased local protest even under no net loss of procedural fairness. Second, even a modest average decrease in legitimacy may be politically consequential if it is concentrated among a most highly engaged, well-resourced faction willing to utilize other veto points. By isolating the effect of process from outcome, this study provides a framework for measuring the extent and intensity of such behavioral consequences in future research.

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Online Appendix

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A Wind Turbine Siting Institutional Reforms

Isolating the relationship between institutions and procedural fairness is highly relevant for energy siting policy globally. Local governments typically control decision-making over new wind facilities, presenting obstacles to development even in countries where wind energy investment is popular, such as Denmark and Sweden (Vasstrøm et al. 2024; Lindvall 2023). Compared to centralized authorities, local legislators are elected and thus incentivized to keep their votes more closely aligned with nearby residents' attitudes. Additionally, local legislators will typically hold more in-person public hearings on wind proposals, facilitating a higher level of face-to-face public input.

In contrast to this local control with high public input, governments across North America and Europe have sought to centralize decision-making authority. Recently, Germany curtailed local control by setting binding top-down wind energy targets and allowing the state and municipal authorities to manage planning and implementation. Additional reforms have removed public hearings for wind proposals unless requested by the developer or local authorities (Uibleisen and Götz 2024). The United Kingdom has similarly exempted large scale wind power installations from local planning approval. In contrast, Norway replaced local control over wind energy development with a national framework in 2009, sparking widespread protest. By 2019, Norway began a series of reforms to strengthen local control over wind siting, illustrating how a loss of legitimacy may threaten long-term policy success (Vasstrøm et al. 2024).

Rather than centralizing authority nationally, regional governments can also preempt local control. Canadian provinces have removed local veto authority, leading to backlash and the punishment of incumbent officials (Stokes 2016). In the U.S., states have also shifted decision-making over utility-scale wind and solar facilities from local governments to state-appointed boards of experts, including Illinois, California, New York, Wisconsin, and Michigan. Other states have had strong state control for much longer, including Connecticut, Maryland, West Virginia, and North Carolina. Another 20 states maintain a hybrid

approach, where large-scale energy facilities and transmission lines are approved at the state level (Chaudhuri and Susskind 2024). While this preemption often requires at least one public hearing, the level of input typically falls short of that gathered through locally elected decision-making bodies.

B Main Experiment Survey Instrument

Consent + Demographics

Standard module of demographic questions and attention checks.

From what you know about global climate change or global warming, which one of the following statements comes closest to your opinion?

- Global climate change has been established as a serious problem, and immediate action is necessary
- There is enough evidence that climate change is taking place and some action should be taken
- We don't know enough about global climate change, and more research is necessary before we take any actions
- Concern about global climate change is exaggerated. No action is necessary
- Global climate change is not occurring; this is not a real issue

In the United States, local governments are in charge of local public services, property taxes, and how land can be used. On a scale from 1 to 7, how often do you agree with the decisions made by your local government? *Response options:* Never; Very rarely; Rarely; Sometimes; Often; Very often; Always; I don't know.

Now, think about your state government. How often do you agree with the decisions made

by your state government? *Response options:* Never; Very rarely; Rarely; Sometimes; Often; Very often; Always; I don't know.

Transition

We will now show you information regarding a potential scenario in your community. After reading the text, please give your opinions about the topic and other current events. Please pay close attention while reading, as you will be asked questions about the content of this scenario after you've read it.

Project Description

A wind energy developer is proposing to build a wind farm on land in your community. They want to build a farm of 60 turbines, each around 340 feet tall, which would power 85,000 homes per year. You would be able to see the wind turbines from your neighborhood.

For context, here's a visual of the turbine height (far right) compared to other known structures:

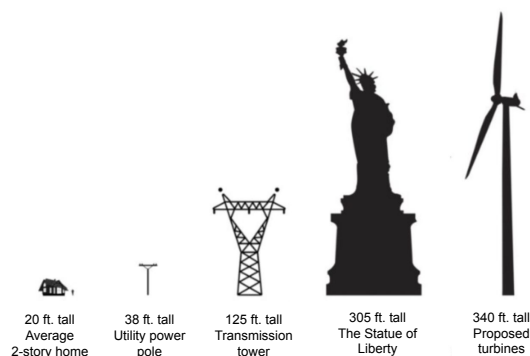


Figure B-1: Visualization of size of each turbine.

Your community is divided. On one hand, some residents are concerned that the facility's size would harm local property values and affect migrating birds. Residents are also concerned about the aesthetics, landscape change, and potential negative impacts on quality of life due to the turbines. Economists have shown that tall turbines like the ones proposed

can lower property values by around 5%.

On the other hand, technical reports estimate that the wind farm would bring over \$2 million per year in new tax revenue for local schools and services and reduce greenhouse gas emissions. Finally, others argue that farmers and landowners should have the right to lease their property to wind developers.

Pretreatment Support for Project

Do you support or oppose the development of this wind farm in your community? *Response options:* Strongly oppose; Oppose; Somewhat oppose; Neither support nor oppose; Somewhat support; Support; Strongly support.

In at least two sentences, why do you support or oppose the wind farm? *Text box answer.*

Treatment 1: Process

According to state law, [your locally elected legislators gets / a state-appointed board of energy experts get] to decide on whether the proposed wind farm will be approved or denied. [Your locally elected legislators / A state-appointed board of energy experts] will [hold a series of local public meetings where residents like you can share your opinions face-to-face with the decision makers / set up an email account where residents like you can send your thoughts, but there will be no public meetings about the proposal].

Expected Outcomes

How likely do you think it is that the wind farm proposal will be approved? *Response options:* Extremely likely; Likely; Somewhat likely; Equally likely and unlikely; Somewhat unlikely; Very unlikely; Extremely unlikely.

Treatment 2: Policy Outcome (included in Branch 2 only)

After much debate, [INSTITUTION] decided to [OUTCOME] the wind farm proposal. The wind farm will [MODIFIER] be built.

- *INSTITUTION: four unique conditions from fully randomizing the decision-making body and level of public input.*
- *OUTCOME: deny / approve*
- *MODIFIER: not / now*

Dependent Variables (both Branch 1 and Branch 2)

Regardless of what you think about the [eventual / NA] decision, do you believe that a vote by [INSTITUTION]—after gathering input from residents through [email / public meetings]—is a fair way to decide on the approval of new wind energy farms?

Response options: Extremely unfair; Unfair; Somewhat unfair; Neither fair nor unfair; Somewhat fair; Fair; Extremely fair.

Thinking about the process and [eventual / NA] decision, do you agree or disagree with each of the following statements? Matrix table, 5-point scale (strongly agree, agree, neither agree nor disagree, disagree, strongly disagree)

- “This decision ought to be accepted and considered the final word on the matter.”
- “People would be justified in protesting this decision.” (*Reverse-coded*). *Note: This item was excluded from the analysis based on protocol specified in the pre-analysis plan. See Section E for discussion. Results are substantively and statistically unchanged regardless of its exclusion.*
- “The body that made this decision had the right to do so.”
- “There ought to be an effort to challenge the decision and get it changed.” (*Reverse-coded*)
- “Even if I personally disagree, I respect the outcome of this process.”

In at least 2 sentences, why do you believe this [eventual / NA] decision [would be / is] either legitimate or illegitimate? *Text box answer*

Placebo Check

In your opinion, how [would/will/would have] this proposed wind farm [affect/affected] property values in your community? *Response options:* 7 - Greatly increase property values; 6; 5; 4 - No effect on property values; 3; 2; 1 - Greatly decrease property values.

Language conditionals:

- “would” - under Branch 1 - outcome not reported
- “will” - under Branch 2 - project approved
- “would have” - under Branch 2 - project rejected

C Turbine Height Implications

Modern onshore wind turbines have an average hub height of 339 feet (Hartman 2024). I interpreted that hub height as a tip of blade height in my visualization of the proposed turbines. How may this misunderstanding have affected the experimental results? My expectation is that taller turbines would lead to an intercept shift downward—less support and less legitimacy overall—but the process conditions would have similar legitimizing treatment effects.

D Descriptive Statistics

The table in this section compares the sample to its relevant population. For demographic traits, I use NHGIS to collect ZIP (ZCTA)-level data from the 2019–2023 5-year American Community Survey (Schroeder et al. 2025). These ZIP code traits are weighted by their population to reflect the composition of residents themselves.

Table D-1: Descriptive statistics: analytic sample ($N = 3,076$) and weighted average of sampled ZIP codes

Characteristic	Survey sample	ZIP population (weighted)
Homeowner (pct.)	77%	72%
Female (pct.)	60%	50%
White, non-Hispanic (pct.)	85%	73%
Income (mean, household)	\$68,761	\$66,233
Age (mean)	50	42
College educated (pct.)	28%	24%
Republican/lean R. (pct.)	54%	NA

Statistics are computed on the analytic sample of 3,076 respondents. The female indicator is coded 1 for respondents who answered “Female” and 0 otherwise; the 19 respondents who selected “Non-binary” or “Other” are therefore coded 0 rather than dropped, so that they remain in the estimation sample for every model. The experiment samples exclusively rural ZIP codes and used demographic quotas based on the ZIP code data. While PureSpectrum has a deeper pool of rural respondents, these quotas were not strict. Still, the sample reflects the rural target population in terms of homeownership, college education, and income. The sample has an over-representation of women, white respondents, and older respondents. The sample is majority (54%) Republican/Republican leaners which is a reasonable target.

The sample exhibits modest deviations from their target populations on several demographic traits, these imbalances are unlikely to bias the estimated treatment effects. The treatments in both experiments are fully randomized at the individual level, and all observed demographic traits are controlled for in the analysis. As a result, differences between the sample and the target population may affect the levels of outcomes but not the internal validity of causal comparisons across treatment conditions (Coppock and McClellan 2019).

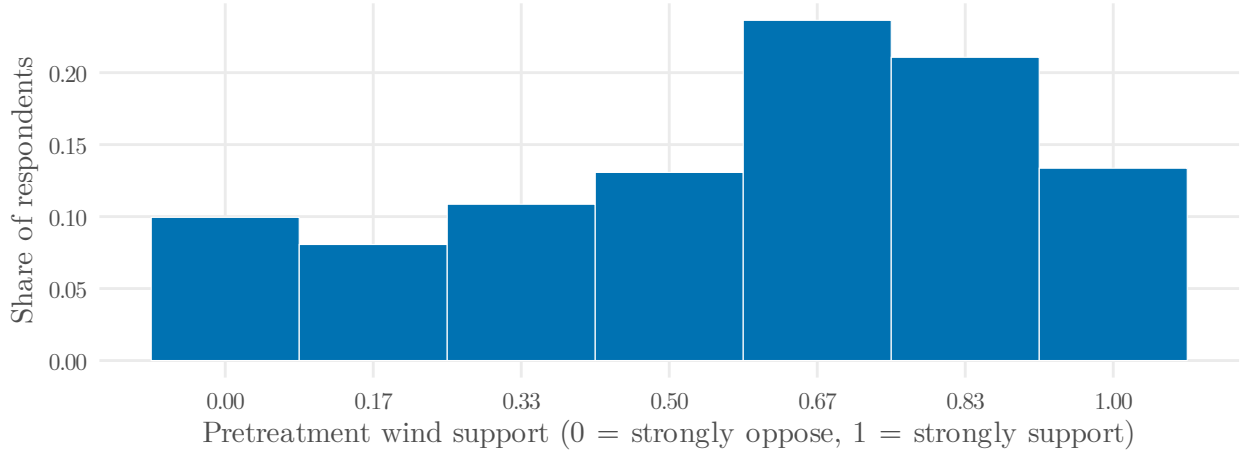


Figure D-2: Distribution of pretreatment support for the wind farm.

E Validation of Legitimacy Scale

The survey instrument included an original five-item legitimacy scale, not the four-item scale reported in the text. The items were as follows, with the order randomized: “Thinking about the process and [eventual / NA] decision, do you agree or disagree with each of the following statements?”

- “This decision ought to be accepted and considered the final word on the matter.”
- “People would be justified in protesting this decision.” (*Reverse-coded*).
- “The body that made this decision had the right to do so.”
- “There ought to be an effort to challenge the decision and get it changed.” (*Reverse-coded*)
- “Even if I personally disagree, I respect the outcome of this process.”

Table E-2: Scale reliability summary

Scale	Alpha raw	Alpha std	Avg item r	G6 smc	S over N
Legitimacy scale	0.69	0.69	0.31	0.69	2.24

Figure E-3 shows the distribution of support for each statement in the legitimacy scale. Support is reverse-coded for the “Protest” and “Change” items. While support for these two statements is lower than for the others, all five items are positively correlated with one

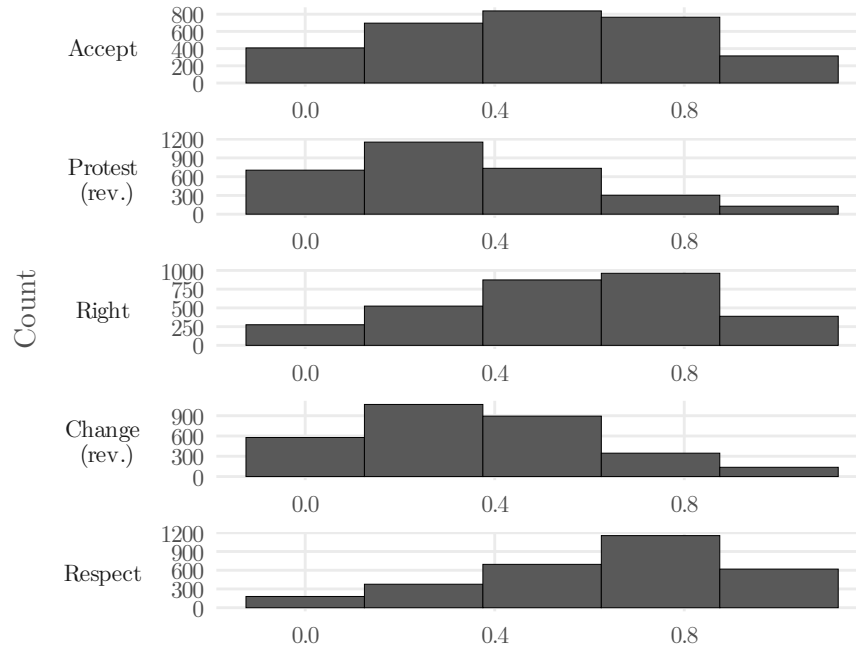


Figure E-3: Histograms of legitimacy scale items. Scaling for items “Protest” and “Change” already reverse-coded to match scale.

another.

The pre-analysis plan stated that items would be dropped to target a Cronbach’s alpha of .70 or higher. With all five items, the Cronbachs’ alpha is .69. Dropping the “Protest” item increases the Cronbach’s alpha to .70. Results are substantively and statistically the same in either approach. However, to strictly adhere to the pre-analysis plan, the manuscript reports estimates using the four-item legitimacy, excluding the “Protest” item.

F Alignment with Local Government

One explanation for the effect of level of decision-making is that respondents may feel poorly represented by their state government. But some respondents may feel poorly represented at both the state and local level. Thus, I asked respondents to evaluate their agreement with decisions by their local government and state government. I subtract agreement with state government with agreement with local government to generate a cleaner measure of preference for local government control.

Figure F-4 shows the interactive effect of process with a preference for local government. For those who prefer local government, a decision made by a state board of experts has a negative effect on fairness and legitimacy, whereas those who prefer state government are essentially unmoved by it. Of course, one’s preference for local government is not randomly assigned, so these respondents may differ in other ways unaccounted for in this design.

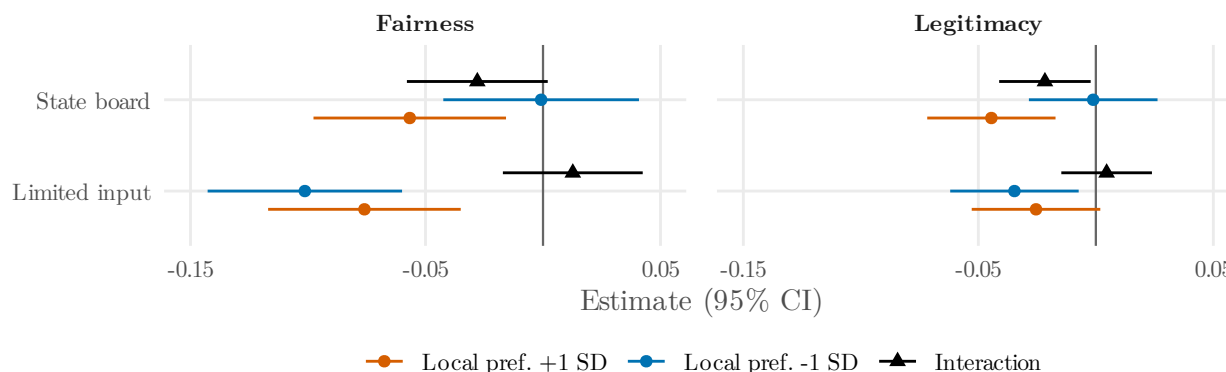


Figure F-4: Interaction between preference for local government and effect of process when the outcome is unknown (Branch 1 only). Point estimates shown with 95% confidence intervals.

G Expected Outcomes

This section reports Research Question 1 of the pre-analysis plan: how does process affect expectations of the outcome? Respondents were asked “How likely do you think it is that the wind farm proposal will be approved?” on a 7-point scale, which I rescale to 0–1.

Following the pre-analysis plan, I regress the expectation item on the two process treatments among respondents in Branch 1. As shown in Figure G-5, neither treatment moves what respondents expect the decision to be. The intervals are narrow enough to rule out anything approaching the size of the process effects on fairness reported in the main text.

The remainder of this section is an exploratory analysis beyond the pre-analysis plan. It pools both branches rather than using Branch 1 alone. Respondents in Branch 2 answered the expectation item before seeing the outcome, so both branches remain usable for it. Figure G-

6 displays the effect of process on whether respondents expected to receive their preferred outcome. The two process treatments cut against each other. On one hand, respondents on average expected that limited public input would decrease their chance of receiving their preferred outcome, which aligns with their evaluations of fairness and legitimacy. On the other hand, the state board treatment actually increased their expectations of receiving their preferred outcome, which cuts against the mechanism of strategic considerations driving fairness and legitimacy.

To be clear, that process affects perceptions of the outcome does not negate the independent effect of process on fairness and legitimacy. Expecting a different outcome is not the same as judging the process through that expectation. Whatever role expectations play before the decision is announced, the Branch 1 versus Branch 2 comparison removes that channel. In Branch 2 there is nothing left to expect, and the process effects there do not attenuate in three out of four models.

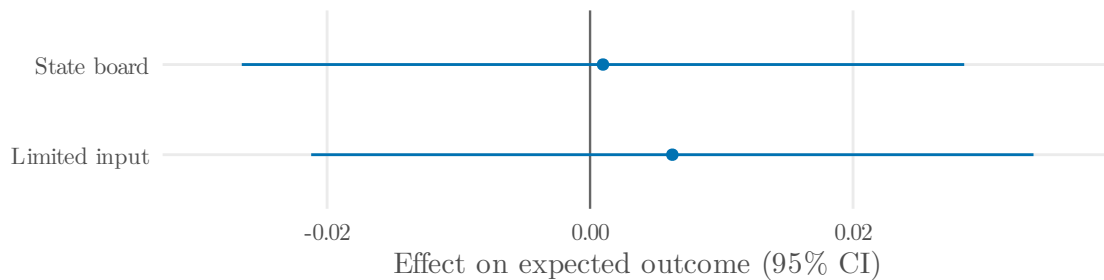


Figure G-5: Effect of process on the expected outcome of the turbine siting decision, as specified in the pre-analysis plan (Branch 1 only). Point estimates shown with 95% confidence intervals.

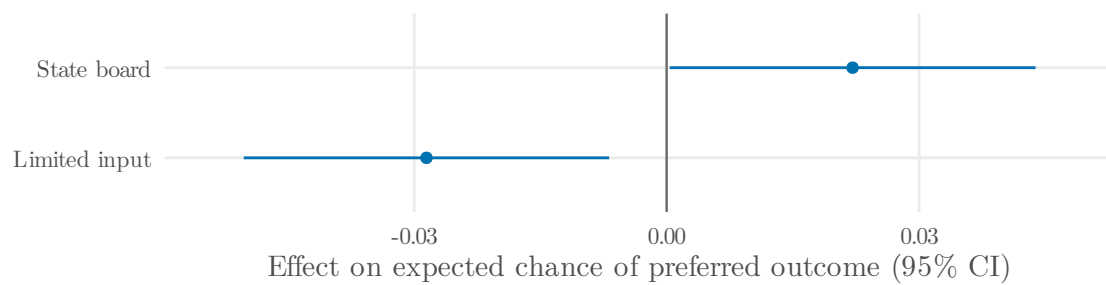


Figure G-6: Effect of process on the expected chance of receiving one's preferred outcome, recoding the expectation item by pretreatment support (expected approval for supporters, one minus expected approval for opponents). Point estimates shown with 95% confidence intervals.

H Results in Tabular Form

H.1 Effect of Process

	Fairness		Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	−0.086*** (0.015)	−0.088*** (0.015)	−0.028** (0.010)	−0.030** (0.010)
State board	−0.032* (0.015)	−0.031* (0.015)	−0.024* (0.010)	−0.024* (0.010)
Homeowner		0.005 (0.018)		−0.021 (0.012)
White, non-Hispanic		0.009 (0.021)		0.022 (0.014)
Income (cont.)		−0.006 (0.003)		−0.001 (0.002)
Female (bi.)		−0.019 (0.015)		−0.003 (0.010)
College or more		0.010 (0.018)		0.013 (0.012)
Age		−0.002*** (0.000)		−0.002*** (0.000)
Republican (−1, 1)		−0.036*** (0.008)		−0.025*** (0.006)
(Intercept)	0.602*** (0.012)	0.744*** (0.030)	0.524*** (0.008)	0.630*** (0.020)
R ²	0.024	0.056	0.009	0.055
Adj. R ²	0.023	0.050	0.008	0.050
Num. obs.	1564	1527	1536	1500
RMSE	0.287	0.284	0.194	0.191

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table H-3: Effect of process on perceived fairness and legitimacy of siting decision (Branch 1 only). Displayed in Figure 3.

	Fairness		Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	−0.087*** (0.014)	−0.089*** (0.015)	−0.029** (0.010)	−0.030** (0.010)
State board	−0.030* (0.014)	−0.029* (0.015)	−0.023* (0.010)	−0.023* (0.010)
Prefer local government	−0.021 (0.012)	−0.017 (0.012)	−0.006 (0.008)	−0.002 (0.008)
Limited input x Local preference	0.018 (0.015)	0.013 (0.015)	0.007 (0.010)	0.005 (0.010)
State board x Local preference	−0.028 (0.015)	−0.028 (0.015)	−0.020 (0.010)	−0.022* (0.010)
Homeowner		0.006 (0.018)		−0.021 (0.012)
White, non-Hispanic		0.009 (0.021)		0.022 (0.014)
Income (cont.)		−0.006 (0.003)		−0.001 (0.002)
Female (bi.)		−0.021 (0.015)		−0.004 (0.010)
College or more		0.013 (0.018)		0.014 (0.012)
Age		−0.002*** (0.000)		−0.002*** (0.000)
Republican (−1, 1)		−0.036*** (0.008)		−0.025*** (0.006)
(Intercept)	0.602*** (0.012)	0.741*** (0.030)	0.524*** (0.008)	0.630*** (0.020)
R ²	0.035	0.065	0.015	0.061
Adj. R ²	0.032	0.058	0.012	0.053
Num. obs.	1563	1526	1535	1499
RMSE	0.286	0.283	0.194	0.190

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table H-4: Effect of process—interacted with preference for local government—on perceived fairness and legitimacy of siting decision (Branch 1 only). Displayed in Figure F-4.

H.2 Resilience of Process

	Fairness		Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	−0.086*** (0.015)	−0.088*** (0.015)	−0.028** (0.010)	−0.029** (0.010)
State board	−0.032* (0.015)	−0.033* (0.015)	−0.024* (0.010)	−0.024* (0.010)
Outcome known	0.020 (0.018)	0.019 (0.018)	0.025* (0.012)	0.026* (0.012)
Limited input x Outcome known	−0.005 (0.021)	−0.006 (0.021)	−0.001 (0.015)	−0.002 (0.015)
State board x Outcome known	−0.003 (0.021)	−0.002 (0.021)	0.010 (0.015)	0.013 (0.015)
Homeowner		−0.003 (0.013)		−0.011 (0.009)
White, non-Hispanic		−0.013 (0.016)		0.002 (0.010)
Income (cont.)		−0.005 (0.002)		−0.001 (0.002)
Female (bi.)		−0.008 (0.011)		−0.002 (0.008)
College or more		0.021 (0.013)		0.019* (0.010)
Age		−0.001* (0.000)		−0.001*** (0.000)
Republican (−1, 1)		−0.029*** (0.006)		−0.015*** (0.004)
(Intercept)	0.602*** (0.012)	0.683*** (0.024)	0.524*** (0.008)	0.584*** (0.016)
R ²	0.025	0.039	0.012	0.026
Adj. R ²	0.024	0.035	0.010	0.022
Num. obs.	3076	3004	3022	2953
RMSE	0.294	0.292	0.204	0.204

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table H-5: Effect of process—interacted with knowledge of decision outcome—on perceived fairness and legitimacy of siting decision (Branches 1 and 2). Displayed in Figure 4.

	Fairness		Legitimacy	
	Model 1	Model 2	Model 3	Model 4
Limited input	−0.087*** (0.016)	−0.088*** (0.016)	−0.030** (0.011)	−0.030** (0.011)
State board	−0.034* (0.016)	−0.034* (0.016)	−0.024* (0.011)	−0.024* (0.011)
Outcome known: preferred	0.108*** (0.022)	0.108*** (0.022)	0.083*** (0.016)	0.084*** (0.017)
Outcome known: not preferred	−0.049* (0.025)	−0.051* (0.025)	−0.030 (0.017)	−0.029 (0.017)
Limited input x Preferred	−0.026 (0.027)	−0.031 (0.027)	0.015 (0.019)	0.013 (0.020)
Limited input x Not preferred	0.007 (0.029)	0.009 (0.029)	−0.010 (0.020)	−0.012 (0.020)
State board x Preferred	0.032 (0.027)	0.035 (0.027)	0.023 (0.019)	0.025 (0.020)
State board x Not preferred	−0.054 (0.029)	−0.054 (0.029)	−0.008 (0.020)	−0.003 (0.020)
Homeowner		−0.012 (0.015)		−0.016 (0.010)
White, non-Hispanic		−0.018 (0.017)		0.003 (0.012)
Income (cont.)		−0.005* (0.003)		−0.001 (0.002)
Female (bi.)		0.002 (0.012)		0.003 (0.008)
College or more		0.024 (0.014)		0.021* (0.010)
Age		−0.001** (0.000)		−0.001*** (0.000)
Republican (−1, 1)		−0.029*** (0.006)		−0.017*** (0.005)
(Intercept)	0.608*** (0.013)	0.703*** (0.026)	0.527*** (0.009)	0.588*** (0.018)
R ²	0.075	0.092	0.069	0.083
Adj. R ²	0.072	0.087	0.066	0.078
Num. obs.	2674	2617	2629	2575
RMSE	0.293	0.291	0.205	0.204

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table H-6: Effect of process—interacted with whether the outcome is unknown, known and preferred, or known and not preferred—on perceived fairness and legitimacy of siting decision (Branches 1 and 2). Displayed in Figure 5.

I One-Sided Test for Power

Table I-7 reports one-sided tests of the full “strategic considerations” null. If respondents value process only because it signals the outcome, then revealing the outcome should eliminate the process effect entirely. Put simply, the interaction should exactly cancel the main effect. I therefore test that sum against zero, one-sided, using its full standard error. Here, rejecting the null means the process effect is resilient to the outcome being revealed. Three out of four models reject at conventional levels. The one exception is the state board effect on legitimacy ($p = .169$), the smallest effect in the study. The failure to reject here may reflect the comparatively limited role of the decision-maker next to public input. Still, the pattern is consistent with sincere procedural values.

Table I-7: One-Sided Tests Against Full Strategic Attenuation

Outcome	Treatment	Estimates			One-Sided Test			
		$\hat{\beta}_1$	$\hat{\beta}_3$	$\hat{\beta}_1 + \hat{\beta}_3$	SE	t	p	Reject H_0
Fairness	Limited Public Input	-0.088	-0.006	-0.094	0.016	-6.021	<0.001	Yes
Fairness	State Board	-0.033	-0.002	-0.035	0.016	-2.213	0.013	Yes
Legitimacy	Limited Public Input	-0.029	-0.002	-0.032	0.011	-2.799	0.003	Yes
Legitimacy	State Board	-0.024	0.013	-0.011	0.011	-0.960	0.169	No

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